



A theremin glove made using Arduino Lilypad (Credit: Arduino Project Hub)

watches and other wearables from being hacked. A good development ecosystem is also necessary for integrating all the necessary features during the development process. It needs to offer all the GUI based tools, firmware, middleware, APIs, and comprehensive documentation. One such development ecosystem is the ModusToolbox from Infineon. It is a collection of libraries and middleware hosted on GitHub, along with an IDE and toolkit that is downloaded and installed on your machine.

PWM and the need for an AFE.

Pulse width modulation (PWM) is used to create a number of effects, including LED dimming, vibration for haptic feedback, and much more. If used in firmware, these strategies call for precise timing and regular CPU attention. Choosing a CPU or controller that supports hardware PWM is crucial.

Second, a lot of sensors used in wearables are analogue sensors, especially those used in biomedical applications that measure heart rate. A unique component, known as the analogue front end (AFE), is necessary for analysing data obtained

from analogue sensors.

Op-amps, filters, and ADCs are used in an AFE to condition and transform the analogue signal into a digital signal that the CPU can handle. At times, some controllers have an AFE integrated into them. Microchip's PIC32CM LS60 family of microcontrollers that were launched in May 2022 actually have ADCs, DACs, and op-amps, which are the features of an inbuilt AFE. Hence,

these controllers are best suited for wearable electronics.

Some trends to look out for

Ever since the development of the first wearable device, the HP-01 wearable calculator, the wearables have come a long way. Some recent trends seen in the wearables are mentioned below.

Open source wearables. The open source revolution has certainly reached wearable electronics—from chips like the Arduino Lilypad to boards to complete wearable projects. Researchers from Portugal have actually developed an open source board based on the low-power, 8-bit, ATtiny20-CCU Microchip AVR microcontroller for the development of wearable and cyber-physical prototypes. Likewise, developmental psychologists at Indiana University and the University of East Anglia have created an open source, wireless vest for measuring autonomic function in infants!

Solar-powered wearables. At times, it makes sense to power wearable devices using sunlight, especially because it ensures that your device would get charged by the sunshine and stay charged. Fitness trackers equipped with solar charg-



E-Tattoo (Credit: University of Texas at Austin and Texas A&M University)